

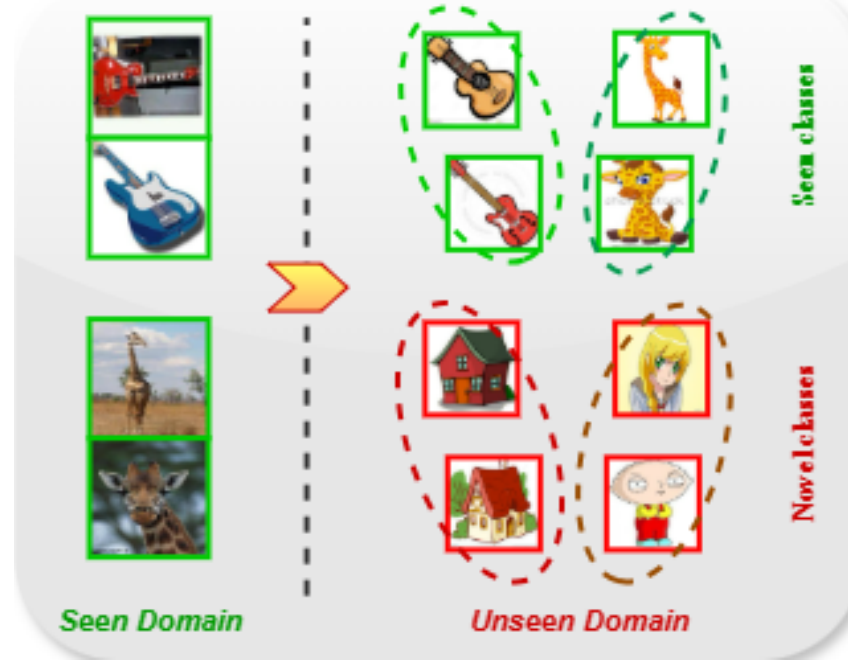
HIDISC: A Hyperbolic Framework for Domain Generalization with Generalized Category Discovery

Vaibhav Rathore, Divyam Gupta, Biplab Banerjee

Indian Institute of Technology Bombay

Introduction and Motivation

- **Domain Generalization for Generalized Category Discovery (DG-GCD)** aims to recognize both **known** and **novel classes** in an **unseen target domain** using only **labeled source data**.
- This setting reflects real-world situations such as **autonomous driving** and **medical diagnosis**, where collecting or labeling target-domain data is often **impractical**.



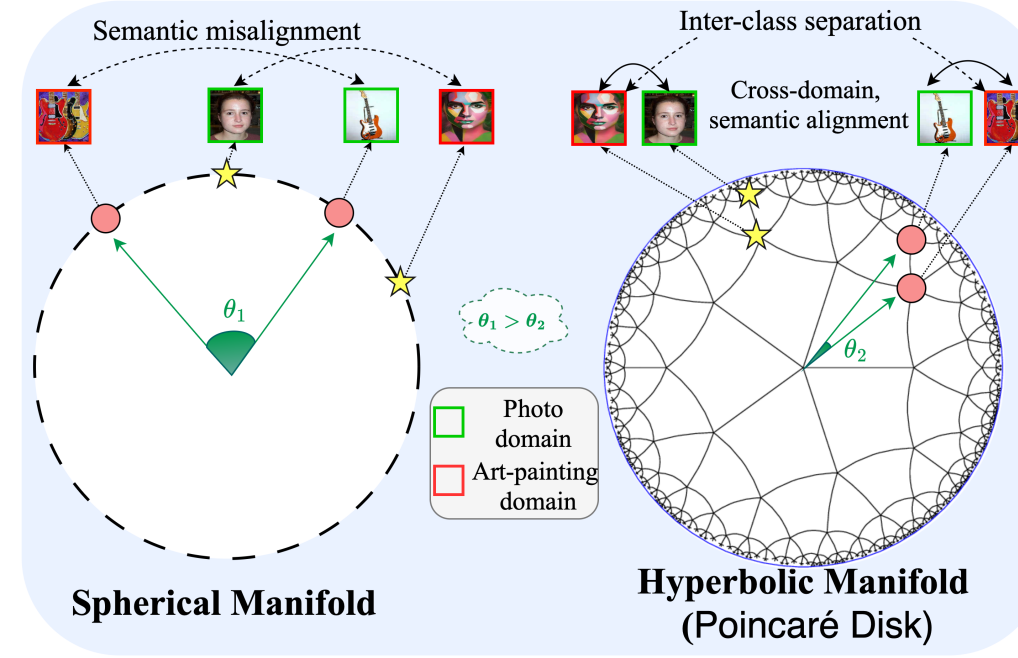
From Euclidean to Hyperbolic Representation Learning

- **Limitations of DG²CD-Net**: The first DG-GCD framework, but relies on **episodic simulation** and **Euclidean embeddings**—which poorly capture semantic hierarchies and lead to high computational cost.
- **Geometric Insight**: Visual data naturally form **hierarchical structures** (e.g., “animal → mammal → dog”). Euclidean space expands linearly and struggles to represent these hierarchies, whereas **hyperbolic space**, with negative curvature, grows exponentially and preserves hierarchy with fewer dimensions.
- **Motivation for HIDISC**: Can hyperbolic geometry provide a unified foundation for solving DG-GCD, addressing both distribution shift and novel-class discovery?

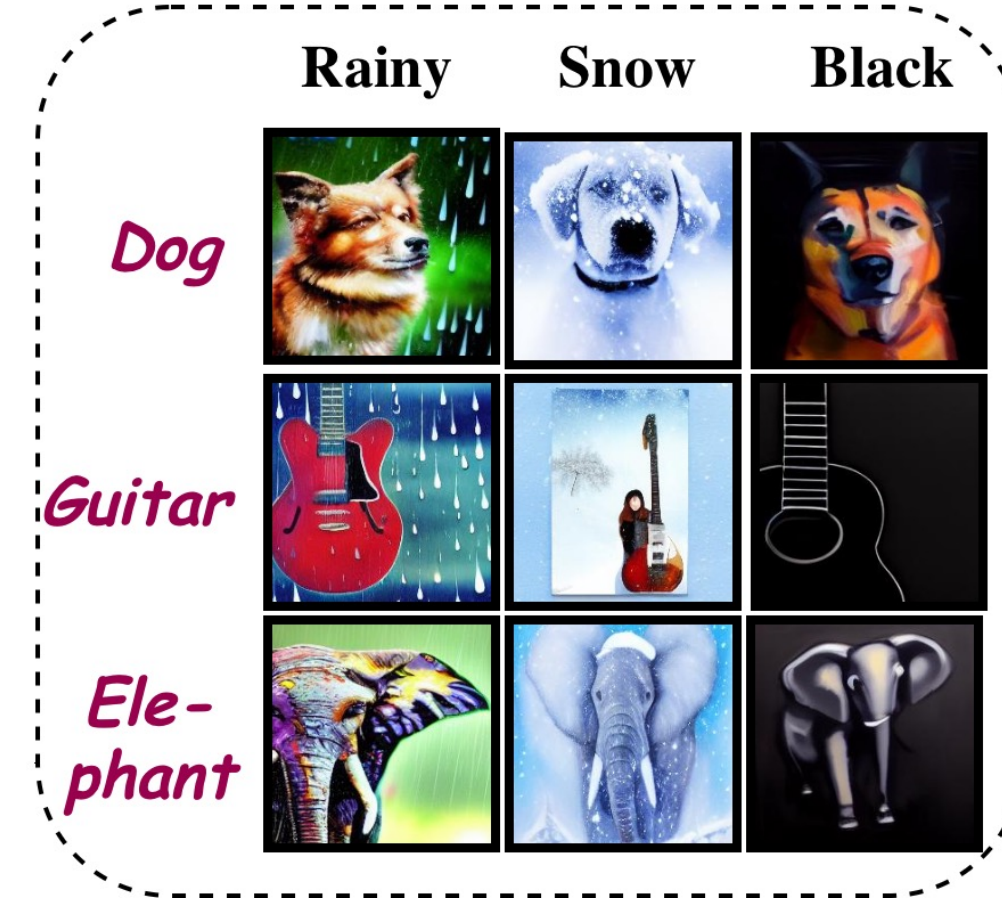
Our Contributions

- **1. Unified Hyperbolic Framework**: We introduce **HIDISC**, the first **hyperbolic-domain representation framework** for DG-GCD that jointly addresses both domain and category shifts—without target data or episodic training.
- **2. Geometry-Aware Learning**: HIDISC integrates **Busemann alignment**, **hyperbolic contrastive regularization**, and **adaptive outlier repulsion** into a single unified loss. Additionally, **Tangent CutMix** augments data in curvature-consistent space for better novel-class generalization.
- **3. Efficiency and Performance**: HIDISC achieves state-of-the-art accuracy on PACS, Office-Home, and DomainNet, while reducing training FLOPs by more than **96×** compared to Euclidean counterparts.

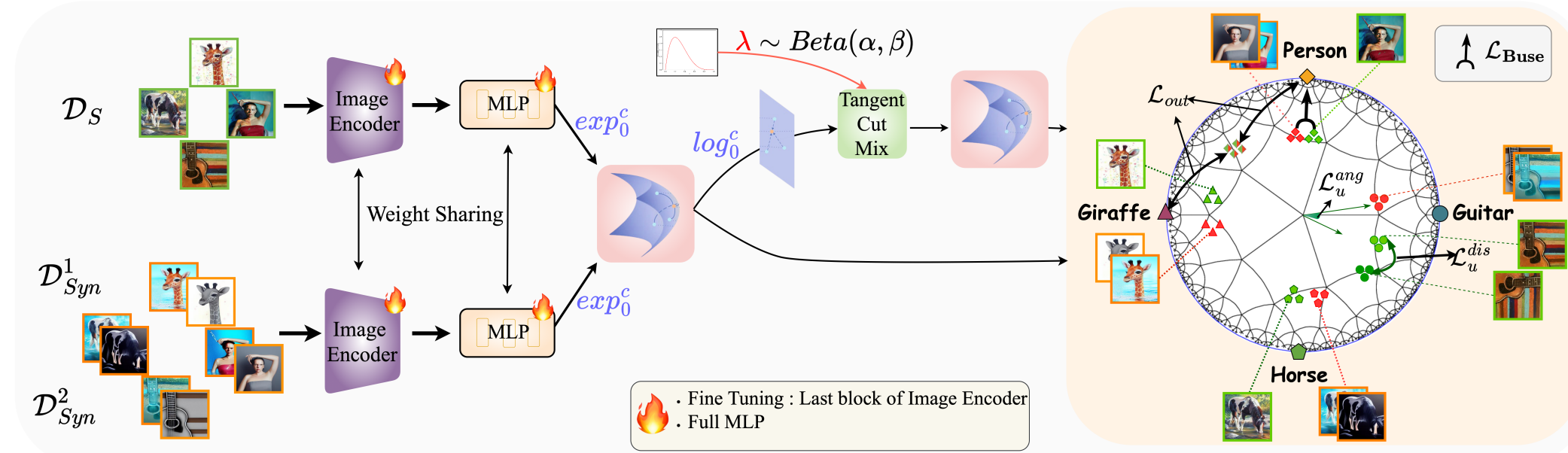
Geometrical Embeddings



Synthetic Images



HIDISC Method



- **1. Penalized Busemann Loss**: Aligns seen-class features to boundary prototypes via hyperbolic distance regularization.

$$\mathcal{L}_{\text{Buse}} = \log \left(\frac{\|z_i - p_{y_i}\|^2}{1 - c\|z_i\|^2} \right) + \phi \log(1 - \|z_i\|^2)$$

- **2. Hybrid Hyperbolic Contrastive Loss**: Enforces local compactness and global separation using both geodesic and angular similarities.

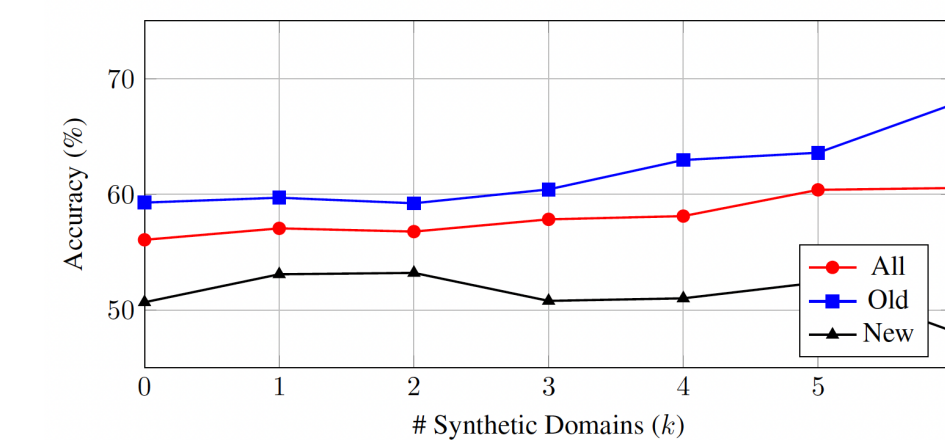
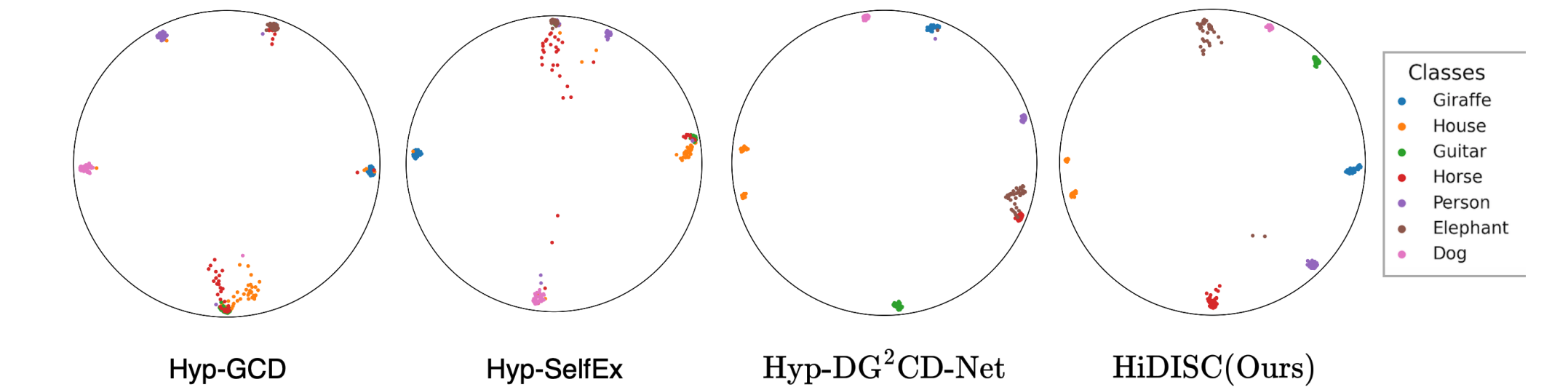
$$\mathcal{L}_u = \frac{1}{|B|} \sum_{i \in B} -\log \frac{\exp(\delta(z'_i, z'_i)/\tau)}{\sum_{j \neq i} \exp(\delta(z'_i, z'_j)/\tau)}, \quad \delta(\cdot, \cdot) = \alpha d[-D_H(\cdot, \cdot)] + (1 - \alpha) \cos(\cdot, \cdot)$$

- **3. Adaptive Outlier Loss**: Repels synthetic (CutMix) samples from known-class regions to encourage open-space generalization.

$$\mathcal{L}_{\text{out}} = \mathbb{E}_{(x,y) \sim \mathcal{P}(\mathcal{D}_{\text{train}})} \sum_{i,j, y_i \neq y_j} \max(0, \gamma - \min_{k \in Y_s} D_H(z_{\text{mix}}^{i,j}, p_k))$$

Results and Ablations

Method	Venue	PACS			Office-Home			DomainNet			Avg.		
		All	Old	New	All	Old	New	All	Old	New	All	Old	New
ViT [60]	ICLR'21	41.98	50.91	33.16	26.17	29.13	21.62	25.35	26.48	22.41	31.17	35.51	25.73
GCD [15]	CVPR'22	52.28	62.20	38.39	52.71	54.19	50.29	27.41	27.88	26.13	44.13	48.09	38.27
SimGCD [25]	ICCV'23	34.55	38.64	30.51	36.32	49.48	13.55	2.84	2.16	3.75	24.57	30.09	15.94
CMS [26]	CVPR'24	28.95	28.13	36.80	10.02	9.66	10.53	2.33	2.40	2.17	13.77	13.40	16.50
SelfEx [64]	ECCV'24	71.82	73.37	71.55	50.18	48.59	52.16	24.78	24.99	24.21	48.93	48.98	49.31
CDAD-Net [18]	CVPR-W'24	69.15	69.40	68.83	53.69	57.07	47.32	24.12	23.99	24.35	48.99	50.15	46.83
GCD+ 6 Synth	CVPR'22	65.33	67.10	64.42	50.50	51.48	48.96	24.71	24.80	21.94	46.85	47.78	45.11
SimGCD+ 6 Synth	ICCV'23	39.76	43.76	35.97	35.57	48.58	12.89	2.71	1.99	4.14	26.01	31.44	17.67
CMS+ 6 Synth	CVPR'24	28.01	26.71	29.04	12.09	12.66	11.13	3.22	3.28	3.03	14.44	14.22	14.40
CDAD+ 6 Synth	CVPR-W'24	60.76	61.67	59.49	53.49	56.90	47.76	23.85	23.88	24.26	46.03	47.47	43.84
Hyp-GCD [22]	CVPR'25	65.33	67.11	64.42	50.13	49.36	48.08	22.88	23.74	25.89	46.12	46.74	46.13
Hyp-SelfEx [64]	ECCV'24	72.44	74.70	71.20	52.91	52.65	52.96	29.30	30.45	26.37	51.55	52.60	50.18
DG ² CD-Net [1] (9 Synth)	CVPR'25	73.30	75.28	72.56	53.86	53.37	54.33	29.01	30.38	25.46	52.06	53.01	50.78
Hyp-DG ² CD-Net [†] (9 Synth)	CVPR'25	74.07	74.40	73.95	49.40	50.29	48.03	22.31	21.52	24.29	48.59	48.74	48.76
HiDISC (Ours) (2 Synth)	—	75.07	75.54	74.52	56.78	59.23	53.21	30.51	31.40	28.41	54.12	55.39	52.05
Δ	—	+1.00	+0.26	+0.57	+2.92	+2.16	-1.12	+1.21	+0.95	+2.04	+2.06	+2.38	+1.27
CDAD-Net (DA) [UB]	CVPR-W'24	83.25	87.58	77.35	67.55	72.42	63.44	70.28	76.46	65.19	73.69	78.82	68.66



Training Domains	Method	All ↑	Old ↑	New ↑
Art	DG ² CD-Net	54.47	53.65	55.54
	HiDISC	54.33	54.75	53.53
Art + Clipart	DG ² CD-Net	66.67	65.78	67.99
	HiDISC	67.34	70.78	62.02
Art + Clipart + Product	DG ² CD-Net	66.83	66.89	66.75
	HiDISC	69.86	79.17	54.24

Config.	$\mathcal{L}_{\text{Buse}}$	$\mathcal{L}_{\text{hrep}}^u$	$\mathcal{L}_{\text{out}}$	Office-Home		
				All	Old	New
Vanilla	✓	✗	✗	26.17	29.13	21.62
$\mathcal{L}_{\text{Buse}}$	✓	✗	✗	56.32	59.74	50.32
$\mathcal{L}_{\text{hrep}}^u$	✗	✓	✗	50.95	49.33	53.06
$\mathcal{L}_{\text{Buse}} + \mathcal{L}_{\text{hrep}}^u$	✓	✓	✗	56.29	60.36	50.41
$\mathcal{L}_{\text{Buse}} + \mathcal{L}_{\text{out}}$	✓	✗	✓	51.04	51.51	50.29
Full HiDISC	✓	✓	✓	56.78	59.23	53.21

Model Variant				Office-Home		
				All	Old	New
- With manual augmentations based $\mathcal{D}_{\text{syn}}$				50.80	51.75	49.15
- Without synthetic domain				56.07	59.29	50.67
- Fixed curvature (c=0.01, close to Euclidean)				56.23	58.65	52.68
- Fixed curvature (c=0.03)				55.67	57.39	52.69
- Cut-Mix (In Euclidean Space)				53.46	54.86	51.06
- Full HiDISC				56.78	59.23	53.21